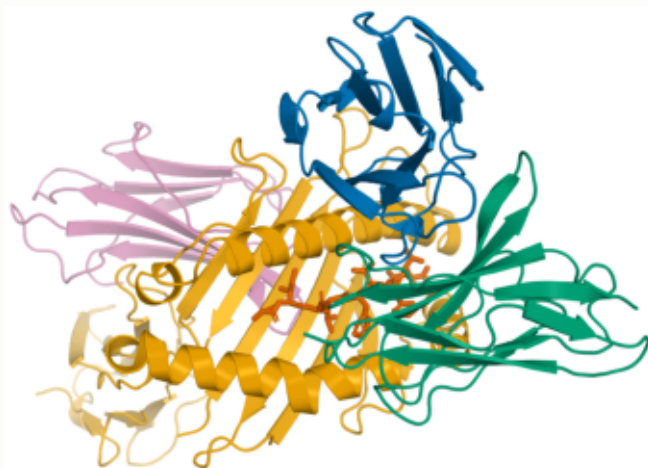


10x TCR repertoire

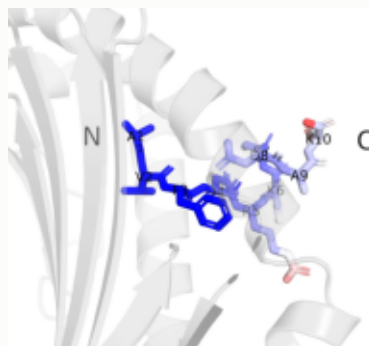


predicted TCR-pMHC
(Protenix, 5 seeds)

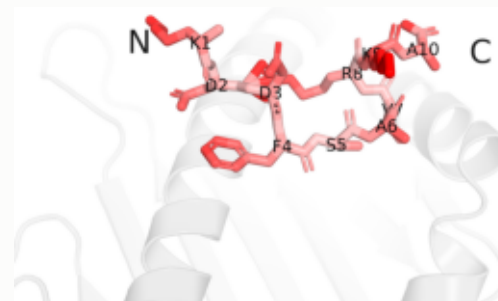
TCR MHC peptide

MHC-peptide presentation

does the peptide bind the HLA?



cognate (high pLDDT)



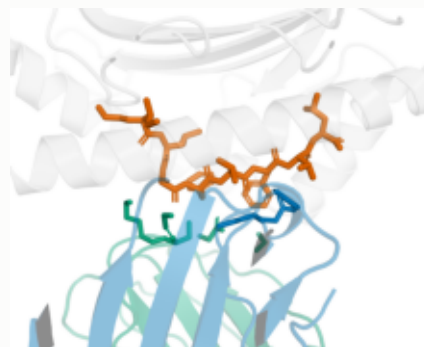
scramble (low pLDDT)



groove confidence
separates binder
from scramble
AUROC up to 0.99

TCR-peptide recognition

does this TCR read the peptide?



pre-registered held-out
retrieval 0.61 (11/18),
 $p = 0.24$: not confirmed

Structural confidence reads presentation, not recognition